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Gabriel's Other Possessions

Melvin Royer

Abstract: Gabriel's Horn is a solid of revolution commonly featured in calculus textbooks as a counter-intuitive example of a solid having finite volume but infinite surface area. Other examples of solids with surprising geometrical finitude relationships have also appeared in the literature. This article cites several intriguing examples (some of fractal type), adds additional ones, and discusses how these topics can enhance a Calculus II or Real Analysis course by adding cohesiveness between topics, challenging students at multiple levels, and illustrating both the power and limitations of a computer algebra system.

Keywords: Gabriel's Horn, solid of revolution, improper integral, volume, surface area.

1. INTRODUCTION

Throughout this article, we consider only solids of revolution generated by revolving about the x -axis a planar region bounded below by the x -axis and above by the graph of a function f , usually taken to be piece-wise continuously differentiable or occasionally piece-wise constant on some (possibly unbounded) interval I of the non-negative x -axis. In a section on improper integrals, most Calculus II textbooks include the example of Gabriel's Horn. Discovered by Evangelista Torricelli in 1641, this solid is formed by rotating about the x -axis the unbounded region in the first quadrant bordered by the graphs of $y = 1/x$, $x = 1$, and the x -axis (see Figure 1).

Gabriel's Horn has finite volume, shown with calculus using the disk method computation

$$V = \int_I \pi [f(x)]^2 dx = \int_1^\infty \pi \left(\frac{1}{x}\right)^2 dx = \pi ,$$

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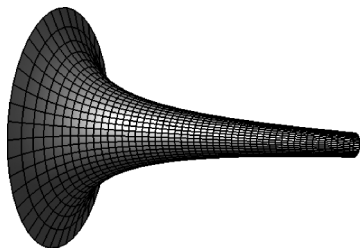


Figure 1. A rendering of Gabriel's Horn.

but infinite surface area, as shown by comparison of the standard integral with a simpler divergent integral:

$$\begin{aligned} A_S &= \int_1^\infty 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx = \int_1^\infty 2\pi \left(\frac{1}{x}\right) \sqrt{1 + \left(-\frac{1}{x^2}\right)^2} dx \\ &\geq \int_1^\infty 2\pi \left(\frac{1}{x}\right) dx = \infty. \end{aligned}$$

Because of these results, it is a common remark today that Gabriel's Horn is a solid which one can "fill but not paint."¹ This remark suggests the converse question of whether there exists a solid that one can "paint but not fill." In other words, if a solid of revolution has finite surface area, must its volume also be finite? In this article we examine three pairs of similar finitude questions.

1. Volume and Surface Area

- (a) Does finite volume imply finite surface area?
- (b) Does finite surface area imply finite volume?

2. Arc Length and Surface Area

- (a) Does finite surface area imply finite arc length of the graph of f ?
- (b) Does finite arc length of the graph of f imply finite surface area?

3. Volume and Planar Area

- (a) Does finite volume imply finite area of the planar generating region?
- (b) Does finite area of the planar generating region imply finite volume?

¹A referee noted that the seeming paradox is a consequence of our selective use of the properties of physical paint. We insist that paint be able to indefinitely flow through the diminishing interior tube of the Horn, yet disallow the possibility of coating the Horn's exterior with a finite quantity of paint whose thickness decreases exponentially as $x \rightarrow \infty$. The nature and possible resolutions of the paradox were discussed in [3]; we continue to use the colloquial terminology throughout the article.

Note that these questions occur in three pairs of converse statements, with the two quantities in each question having some geometrical relationship. The surface “contains” the solid, the arc “generates” the surface under rotation, and the planar region “generates” the solid under rotation. Hence, some finitude relationships might intuitively be expected.

Sections 2, 3, and 4 of the article answer each of these pairs of questions in turn. Along the way, we make several connections between the improper integrals needed to investigate the case when f is a continuously differentiable function with the infinite series needed in the case when f is a related piece-wise constant function. These connections are made to emphasize the relationship between convergence of improper integrals and convergence of infinite series, which can be enlightening to calculus students. Section 5 summarizes the results including, for the sake of completeness, the remaining possible finitude relationships between quantities whose geometrical connection is less apparent. Section 6 then turns to more exotic cases when the function f is mildly pathological in a fractal-like sense. Finally, in Section 7 we discuss the pedagogical value of using such examples in the classroom and give suggestions for doing so.

2. VOLUME AND SURFACE AREA

The question “does finite volume imply finite surface area?” is immediately answered in the negative by Gabriel’s Horn. Another example, dubbed Gabriel’s Wedding Cake by its discoverer, Julian Fleron, illustrates the same point [2]. The Wedding Cake is generated by composing the floor function with that of Gabriel’s Horn, resulting in $f(x) = \frac{1}{\lfloor x \rfloor}$ on the interval $1 \leq x$ (see Figure 2).

The expression for the total volume of the cylindrical layers, $V = \sum_{n=1}^{\infty} \pi \left(\frac{1}{n}\right)^2 (1)$, leads to a convergent p -series. In fact, Euler’s result $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ (or a computer algebra system if this result is unfamiliar) shows that the Wedding Cake has volume $\frac{\pi^3}{6}$. The surface is composed of lateral sides and annular tops for each layer, and its area A_S can be computed as follows:

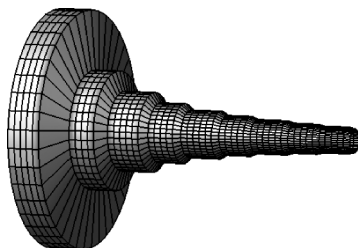


Figure 2. A rendering of Gabriel’s Wedding Cake.

$$A_S = \sum_{n=1}^{\infty} 2\pi \left(\frac{1}{n}\right) (1) + \sum_{n=1}^{\infty} \pi \left[\left(\frac{1}{n}\right)^2 - \left(\frac{1}{n+1}\right)^2 \right].$$

Although the latter telescoping series converges to π , the former series is a multiple of the divergent harmonic series. Gabriel's Wedding Cake thus has finite volume but infinite surface area, as does Gabriel's Horn. The Wedding Cake is therefore another "fill but not paint" solid, or as Fleron notes, "eat but not frost."

We next turn to the question "does finite surface area imply finite volume?" In contrast to the above, the answer for the case when f is continuously differentiable, on some interval I turns out to be "yes." Following van Maanen [6], first note that f must be bounded on I for the surface area to be finite (since the area of the surface must be at least as large as the area of its projection onto any plane perpendicular to the axis of revolution). But given a number M for which $|f(x)| \leq M$ everywhere on I , the volume is bounded by a multiple of the surface area:

$$\begin{aligned} V &= \int_I \pi [f(x)]^2 dx \leq \int_I \pi M |f(x)| dx \leq \int_I \pi M |f(x)| \sqrt{1 + [f'(x)]^2} dx \\ &= \left(\frac{M}{2}\right) A_S. \end{aligned}$$

3. ARC LENGTH AND SURFACE AREA

The next pair of questions deals with the relationship between the arc length of the graph of f and the area of the surface generated by rotating this curve about the x -axis. Regarding the question, "does finite surface area imply finite arc length of the graph of f ?", a solid with similar appearance to Gabriel's Horn, which we name Gabriel's Funnel, serves as a counter-example. Let $f(x) = \frac{1}{x^2}$ on $1 \leq x$. Then the arc length and surface area of the Funnel are given by

$$L = \int_1^{\infty} \sqrt{1 + [f'(x)]^2} dx = \int_1^{\infty} \sqrt{1 + \left(-\frac{2}{x^3}\right)^2} dx \geq \int_1^{\infty} dx = \infty$$

and

$$A_S = \int_1^{\infty} 2\pi \left(\frac{1}{x^2}\right) \sqrt{1 + \left(-\frac{2}{x^3}\right)^2} dx \leq \int_1^{\infty} \frac{2\pi}{x^2} \sqrt{5} dx < \infty.$$

The answer to the question is thus "no." If we imagine one person attempting to hike along the graph of f while someone else tries to paint the surface, we can say that Gabriel's Funnel is a "paint but not hike" solid.

Next, consider the question “does finite arc length of the graph of f imply finite surface area?” First note that, if f is continuously differentiable with finite arc length L , then f must be bounded in magnitude by some constant M . But then its surface area

$$\begin{aligned} A_S &= \int_I 2\pi |f(x)| \sqrt{1 + [f'(x)]^2} dx \leq \int_I 2\pi M \sqrt{1 + [f'(x)]^2} dx \\ &= (2\pi M) L \end{aligned}$$

must also be finite, so the answer to the question is “yes.” In other words, a smooth “hike but not paint” solid cannot exist.

Note the geometrical analogy between Questions 1 and 2 — finitude of the lower dimensional quantity implies finitude of the quantity of higher dimension, but the converse is not true.

4. VOLUME AND PLANAR AREA

The question “does finite volume imply finite area of the planar generating region?” is readily answered. Computing the area of the region generating Gabriel’s Horn, $A = \int_1^\infty \frac{1}{x} dx = \infty$, immediately shows that the answer is “no.” One physical interpretation of this result is that even though Gabriel’s Horn has finite volume, attempting to bisect the Horn with a plane containing the x -axis would require cutting through an infinite cross-sectional area of gold (a likely metal for construction of angel horns). Thus, Gabriel’s Horn can be called a “fill but not slice” solid.

Finally, consider the question “does finite area of the planar generating region imply finite volume?” If f were bounded in magnitude by some number M on I , then the volume is given by

$$V = \int_I \pi [f(x)]^2 dx \leq \int_I \pi M |f(x)| dx = (\pi M) A ,$$

which would show that finite planar area does indeed imply finite volume. But, in this case, f need not be bounded. Consider the example of John Dawson Jr., dubbed Thor’s Anvil [1]. Let $f(x) = \frac{1}{\sqrt{x}}$ on the interval $0 < x \leq 1$. (Note that this function is the inverse of the one generating Gabriel’s Funnel). Presumably the Anvil is known to be Thor’s, since its base spreads to cover the entire yz -plane and the blacksmith must therefore stand at infinity when using the flat surface (i.e., the unit disk in the plane $x = 1$ centered at $(x, y, z) = (1, 0, 0)$) (see Figure 3).

Simple computations for the volume and planar area show $A = \int_0^1 \frac{1}{\sqrt{x}} dx = 2$, while $V = \int_0^1 \pi \left(\frac{1}{\sqrt{x}}\right)^2 dx = \infty$. Thus, Thor’s Anvil is

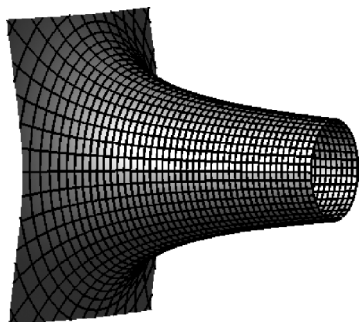


Figure 3. A rendering of Thor's Anvil.

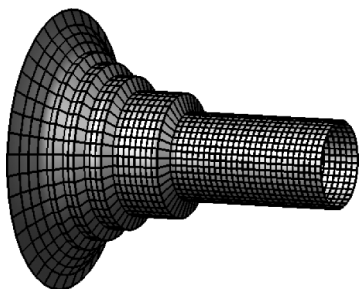


Figure 4. A rendering of Gabriel's Tower of Hanoi.

a solid, having cross-sections of finite planar area, but having infinite volume (i.e., a “slice but not fill” solid).

Another such example, applying the floor function to the function of Thor's Anvil, is Gabriel's Tower of Hanoi, generated from the function $f(x) = \left(\left\lfloor \frac{1}{x} \right\rfloor\right)^{1/2}$ on the interval $0 < x \leq 1$ (see Figure 4).

This Tower, composed of an infinite number of disks of increasing radius and decreasing thickness as $x \rightarrow 0^+$, has planar area and volume,

$$A = \sum_{n=1}^{\infty} \sqrt{n} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}(n+1)} \leq \sum_{n=1}^{\infty} \frac{1}{n^{3/2}} < \infty$$

and

$$V = \sum_{n=1}^{\infty} \pi (\sqrt{n})^2 \left(\frac{1}{n} - \frac{1}{n+1} \right) = \sum_{n=1}^{\infty} \frac{\pi}{n+1} \geq \sum_{n=1}^{\infty} \frac{\pi}{2n} = \infty,$$

and, thus, Gabriel's Tower of Hanoi is also a “slice but not fill” solid.

We have thus concluded that the finitude of the volume of a solid and the finitude of the area of its planar generating region are independent of each other.

5. SUMMARY OF RELATIONSHIPS

We are considering finitude relationships among the arc length L , planar area A , surface area A_S , and volume V . Specifically, we have examined the pairs (V, A_S) , (L, A_S) , and (V, A) . Of the three remaining pairs, (A, A_S) and (L, V) have less obvious geometrical connections, while (A, L) is a known relationship in the plane. For completeness, we summarize results from all six relationships. The only additional computation needed is the following, which directly shows that finite surface area implies finite planar area:

$$A = \int_I f(x) dx \leq \int_I 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx = A_S$$

Now let $\mathbf{F}$ denote the set of all piece-wise continuously differentiable functions of the form $f : I \rightarrow \Re$ where I is some interval in $\Re$. Define $\mathbf{L} = \{f \in \mathbf{F} : L \text{ is finite}\}$, with analogous definitions for the sets $\mathbf{A}$, $\mathbf{A}_S$, and $\mathbf{V}$. These definitions allow us, for example, to represent the finitude relationship in the previous paragraph as the set containment $\mathbf{A}_S \subseteq \mathbf{A}$. The Venn diagram in Figure 5 then illustrates all possible finitude relationships. Note that the sphere and infinite cylinder were included as solids of revolution to illustrate that all regions of the diagram are in fact non-empty.

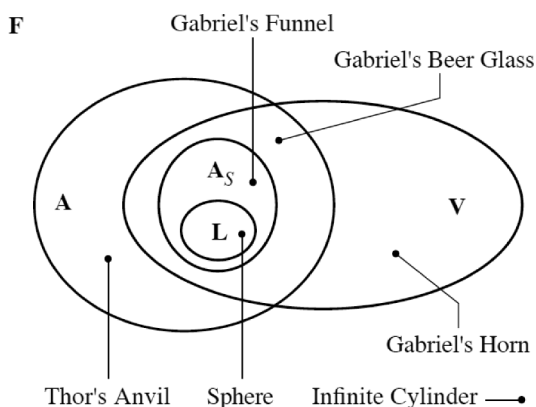


Figure 5. A Visualization of all possible Finitude Inclusion Relationships.

6. FRACTAL-LIKE EXAMPLES

The finitude counter-examples considered thus far all involve solids that are unbounded in $\mathbb{R}^3$. A natural conjecture would perhaps be that such counter-examples inherently depend on unboundedness. This conjecture is certainly true for both parts of Question 3, where boundedness of the solid ensures finitude of both geometrical measurements involved. Consider, however, the Supersine Solid, reported by William Love [5], formed by rotating the region bounded above by $f(x) = \left|\sin \frac{1}{x}\right|$ on $0 < x \leq \frac{2}{\pi}$ about the x -axis (see the imperfectly rendered graph in Figure 6).

This solid clearly has finite volume, being bounded in $\mathbb{R}^3$, but its surface area is given by

$$\begin{aligned} A_S &= \int_0^{\frac{2}{\pi}} 2\pi \left| \sin \frac{1}{x} \right| \sqrt{1 + \left(\frac{-\cos(1/x)}{x^2} \right)^2} dx \\ &\geq \int_0^{\frac{2}{\pi}} 2\pi \left| \sin \frac{1}{x} \right| \left(\frac{|\cos(1/x)|}{x^2} \right) dx . \end{aligned}$$

Using the trigonometric identity $2 \sin \alpha \cos \alpha = \sin 2\alpha$, followed by the substitution $u = \frac{1}{x}$ shows

$$A_S \geq \pi \int_0^{\frac{2}{\pi}} \frac{|\sin(2/x)|}{x^2} dx = \pi \int_{\frac{\pi}{2}}^{\infty} |\sin 2u| du .$$

The latter integral diverges (the integrand is periodic and almost everywhere positive) and thus the Supersine Solid is a bounded “fill but not paint” solid. Note that the surface area could also be shown finite by simply observing that each of the Solid’s lobes projects to the unit disk in the yz -plane, and thus the surface area is at least as large as the area of infinitely many unit disks.

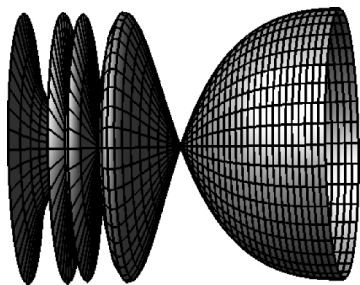


Figure 6. A rendering of the Supersine Solid.

The Supersine Solid achieves boundedness but is geometrically complex. Since its lobes are connected only at a single point, for example, it could not be physically filled by pouring in paint from the top. Kifowit and Stamps [4] propose a less pathological “fill but not paint” example called Gabriel’s Beer Glass (see Figure 7), defined by revolving about the x -axis the region bounded above by the graph of

$$f(x) = \begin{cases} 2 + x \cos \frac{\pi}{x}, & 0 < x \leq 2 \\ 2, & x = 0 \end{cases}.$$

This function is continuous on $0 \leq x \leq 2$ and continuously differentiable on $0 < x \leq 2$.

Being bounded, the solid has finite volume. To show that the surface area is infinite, we bound the surface area from below by the area of the projection of the surface onto the yz -plane. As shown in Figure 8, Gabriel’s Beer Glass includes sections $S_1, S_2, \dots$, where S_n is that part of the glass between $x = \frac{1}{2n+2}$ and $x = \frac{1}{2n}$. The annulus A_n , with inner radius 2 and outer radius $2 + \frac{1}{2n}$, is a subset of the yz -plane projection of S_n . The surface area of just the section S_n is

$$A_S(S_n) \geq \text{area}(A_n) = \pi \left(\left[2 + \frac{1}{2n} \right]^2 - 2^2 \right) > \frac{2\pi}{n},$$

and therefore summing the surface areas of all the sections shows that the total surface area of the Beer Glass is infinite by comparison to the harmonic series.

For a bounded example of a “paint but not hike” solid, consider Gabriel’s Lava Lamp (see Figure 9), formed by revolving about the x -axis the region bounded above by the function

$$f(x) = \begin{cases} x \cos^2 \frac{1}{x}, & 0 < x \leq 1 \\ 0, & x = 0 \end{cases}.$$

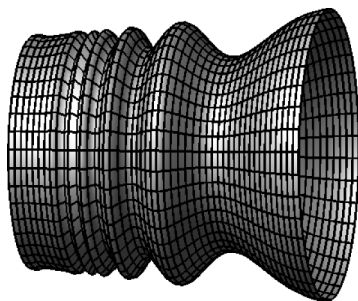


Figure 7. A rendering of Gabriel’s Beer Glass.

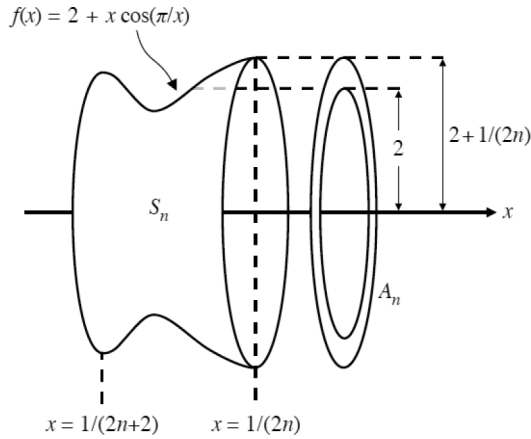


Figure 8. A diagram of the sections of Gabriel's Beer Glass.

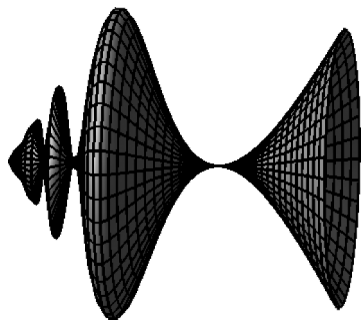


Figure 9. A rendering of Gabriel's Lava Lamp

The Lava Lamp solid is clearly bounded and has surface area

$$\begin{aligned} A_S &= \int_0^1 2\pi x \cos^2 \frac{1}{x} \sqrt{1 + \left(\cos^2 \frac{1}{x} + \frac{2}{x} \sin \frac{1}{x} \cos \frac{1}{x} \right)^2} dx \\ &= \int_0^1 2\pi \cos^2 \frac{1}{x} \sqrt{x^2 + \left(x \cos^2 \frac{1}{x} + 2 \sin \frac{1}{x} \cos \frac{1}{x} \right)^2} dx, \end{aligned}$$

which converges, since the integrand is bounded and non-negative. To show that the arc length is infinite, note that as pictured in Figure 10, the generating curve of the Lava Lamp includes the arcs $L_1, L_2, \dots$, where L_n is that portion of the curve between $x = \frac{1}{n\pi + \pi/2}$ and $x = \frac{1}{n\pi - \pi/2}$. The maximum value of $f(x) = x \cos^2(1/x)$ on this interval must be at least (it is actually greater than)

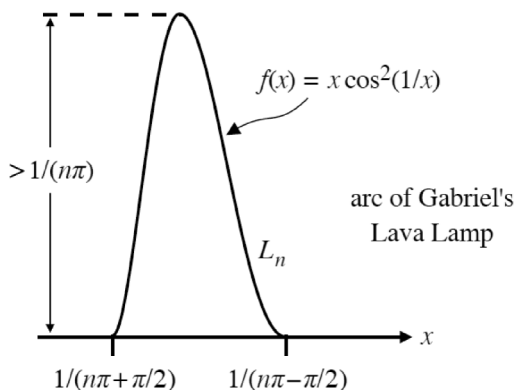


Figure 10. The arc of Gabriel's Lava Lamp.

$f\left(\frac{1}{n\pi}\right) = \frac{1}{n\pi}$. The fact that the arc length of a curve is no less than the length of its projection onto the y -axis shows that L_n has arc length at least $\frac{1}{n\pi}$. The total arc length of the Lava Lamp is therefore infinite by comparison to the geometric series.

7. PEDAGOGICAL VALUE

Analyzing Gabriel's equipment seems a good student activity pedagogically for several reasons. First, even though the questions are readily understandable and the graphs visually appealing, the results are mostly unpredictable, and students must work carefully to get correct answers. Second, the difficulty of the possible examples varies widely. Gabriel's Horn, for example, can be easily analyzed by nearly any Calculus II student, whereas Gabriel's Beer Glass makes a reasonably challenging Real Analysis exercise, especially if students attempt an algebraic approach such as the one given in the Appendix. Third, by including some examples such as Gabriel's Wedding Cake, conceptual connections are made between convergence of infinite series and that of improper integrals (a connection that may be overlooked in a Calculus II course).

Fourth, this analysis is a very good example of a situation where technology is nearly necessary, but far from sufficient. In addition to generating the graphs, a computer algebra system becomes very helpful for the more complicated cases in merely setting up the necessary integrals. The *Maple* code, which is shown in Figure 11 along with its associated output in Figure 12, makes experimentation very easy for the student.

However, for many of the surface area and arc length calculations, the computer is unable to evaluate the integrals or even determine their finitude as it did in Figure 12. In such a case, the student must carefully choose either

```

with(Student[Calculus1]):
f:=x->1/x;
a:=1; b:=infinity; bgraph:=10;
'Volume',VolumeOfRevolution(f(x),x=a..b,output=integral)
      =VolumeOfRevolution(f(x),x=a..b);
'Surface Area',SurfaceOfRevolution(f(x),x=a..b,
      output=integral)=SurfaceOfRevolution(f(x),x=a..b);
plot(f(x),x=a..bgraph,view=[0..10,-.1..1.5],filled=true);
VolumeOfRevolution(f(x),x=a..bgraph,output=plot);

```

Figure 11. The Maple Code.

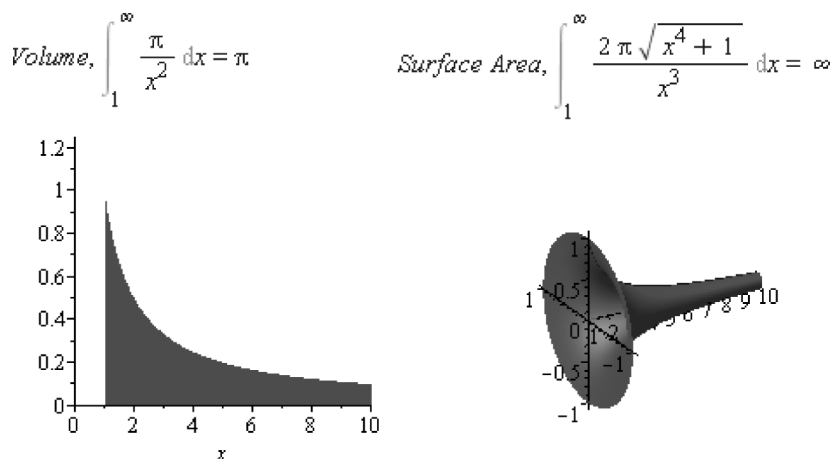


Figure 12. Output of Maple Code.

a geometrical argument or appropriate comparison functions when applying comparison tests for the improper integrals. Deciding when to use technology, how to intermingle geometrical and algebraic techniques, and how to identify negligible terms of an integrand are valuable preparatory exercises for further study in analysis. In obtaining the following inequality below from the Supersine Solid, for example,

$$\begin{aligned}
 A_S &= \int_0^{\frac{2}{\pi}} 2\pi \left| \sin \frac{1}{x} \right| \sqrt{1 + \left(\frac{\cos(1/x)}{x^2} \right)^2} dx \\
 &\geq \int_0^{\frac{2}{\pi}} 2\pi \left| \sin \frac{1}{x} \right| \left(\frac{|\cos(1/x)|}{x^2} \right) dx,
 \end{aligned}$$

the student must identify that the surface area contribution from the first term under the radical is negligible compared to that from the second as $x \rightarrow 0^+$.

The author has used some of these examples as a group project activity in a Calculus II course. The *Maple* code from Figure 11 was provided, as were hints for applying comparison tests. For examples such as Gabriel's Beer Glass, rigorous proof was not required. Instead, students were asked to identify dominant terms and make educated guesses about finitude. As expected, this activity was more successful with stronger calculus students, and parts might be more appropriate for a Real Analysis course.

ACKNOWLEDGMENT

The author would like to thank the referees for many helpful comments, especially for suggesting simpler geometrical arguments to replace some algebraic inequalities, and the Venn diagram in Figure 5 to cleanly illustrate all possible relationships.

APPENDIX

For purposes of illustrating the relative level of complexity, an algebraic proof that Gabriel's Beer Glass has infinite surface area follows. The surface area is

$$\begin{aligned} A_S &= \int_0^2 2\pi \left(2 + x \cos \frac{\pi}{x}\right) \sqrt{1 + \left(\frac{\pi}{x} \sin \frac{\pi}{x} + \cos \frac{\pi}{x}\right)^2} dx \\ &= \int_0^2 2\pi \left(2 + x \cos \frac{\pi}{x}\right) \sqrt{1 + \left(\frac{\pi}{x} \sin \frac{\pi}{x}\right)^2 + \left(\frac{2\pi}{x}\right) \sin \frac{\pi}{x} \cos \frac{\pi}{x} + \cos^2 \frac{\pi}{x}} dx. \end{aligned}$$

To show that this integral diverges, let $J = \bigcup_{n=1}^{\infty} \left[\frac{1}{n+1/2}, \frac{1}{n+1/6}\right]$. Note that if $x \in J$ then:

1. $x \in (0, 1)$ and thus $2 + x \cos \frac{\pi}{x} \geq 1$,
2. $\frac{1}{j+1/2} \leq x \leq \frac{1}{j+1/6}$ for some positive integer j . Consequently,
 - (a) $\pi + 2\pi j \geq \frac{2\pi}{x} \geq \frac{\pi}{3} + 2\pi j$, and thus $2 \sin \frac{\pi}{x} \cos \frac{\pi}{x} = \sin \frac{2\pi}{x} \geq 0$, and
 - (b) $\frac{\pi}{2} + \pi j \geq \frac{\pi}{x} \geq \frac{\pi}{6} + \pi j$, and thus $\frac{1}{2} \leq \left|\sin \frac{\pi}{x}\right|$.

Using (1), (2a), and (2b) in that order shows

$$A_S \geq \int_J 2\pi \sqrt{1 + \left(\frac{\pi}{x} \sin \frac{\pi}{x}\right)^2 + \left(\frac{2\pi}{x}\right) \sin \frac{\pi}{x} \cos \frac{\pi}{x} + \cos^2 \frac{\pi}{x}} dx$$

$$\begin{aligned}
&\geq 2\pi \int_J \frac{\pi}{x} \left| \sin \frac{\pi}{x} \right| dx \\
&\geq \pi^2 \int_J \frac{1}{x} dx \\
&= \pi^2 \sum_{n=1}^{\infty} \int_{1/(n+1/2)}^{1/(n+1/6)} \frac{1}{x} dx \\
&= \pi^2 \sum_{n=1}^{\infty} \ln \left(\frac{n+1/2}{n+1/6} \right)
\end{aligned}$$

and the latter series diverges by the Integral Test.

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BIOGRAPHICAL SKETCH

Melvin Royer is an Associate Professor of Mathematics and Division Chair at Indiana Wesleyan University, where he has taught for the past eight years. His Ph.D. is from Purdue University, in the area of partial differential equations. His current professional interests include dynamical systems, mathematical biology, and classroom projects and demonstrations to enhance the teaching of analysis.